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ENGINEERING**

Final Design Report
**Bleeding Control Simulator: Real-Time Hemorrhage Training
System**

MECH 4025

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Abstract

This report presents the design, development, and validation of a Bleeding Control Simulator created for BraveHeart First Aid to enhance the realism and effectiveness of hemorrhage control training. The system integrates a fluid flow network, medical prosthetic, embedded sensing components, and a real-time feedback interface to simulate a range of bleeding scenarios while quantitatively evaluating user performance.

The simulator is capable of producing adjustable volumetric flow rates and pressures representative of mild to severe bleeding conditions. A diaphragm pump-driven hydraulic system circulates a cost-effective synthetic blood analogue through a prosthetic limb, while thin-film pressure sensors, a pressure transducer, and a flow sensor capture user interaction and system response. These data are processed through an Arduino-based acquisition system and visualized using a MATLAB graphical interface, providing immediate and intuitive feedback through both on-screen displays and LED indicators.

Design decisions were guided by performance requirements, cost constraints, and usability considerations, with iterative prototyping used to refine subsystem integration. Experimental validation confirmed that the system meets or exceeds key functional requirements, including flow control, pressure measurement, and simulation duration, while maintaining a total cost of \$1,326.21 and a blood substitute cost of \$0.51 per litre.

The final prototype demonstrates a portable, scalable, and cost-effective training solution capable of improving the quality of first aid education. Supporting documentation, testing procedures, and operational guidelines are provided to ensure successful long-term implementation and future development by the client.

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1. Background and Design Requirements

Each year, between forty and eighty Nova Scotians die from bleeding-related injuries. Approximately 15% of these deaths are considered preventable, largely due to delays or the improper implementation of bleeding control techniques [13]. Many of these fatalities can be attributed to ineffective bleeding interventions immediately following injury, underscoring the critical role of first responders in emergency medical situations. However, responders may be uncertain about how to intervene appropriately, particularly when they are untrained bystanders rather than medical professionals. The unpredictable timing and severity of medical emergencies highlight the importance of first aid training in both the workplace and everyday life, as it provides individuals with the knowledge and skills necessary to save lives.

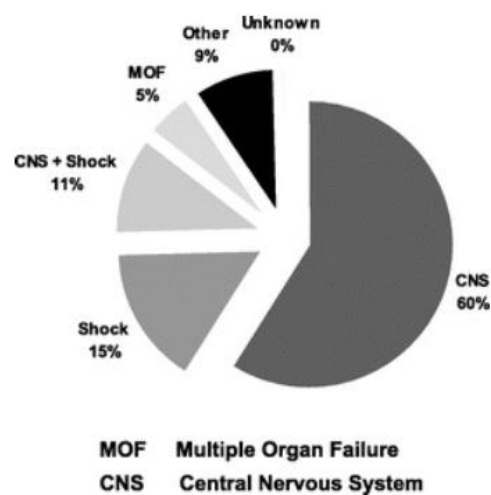


Figure 1: Causes of death at a level 1 trauma center. Plot taken from [13].

First aid training is commonly delivered by private organizations that provide hands-on opportunities to learn and apply essential techniques. Although realistic training scenarios produce the most effective outcomes, many training providers continue to rely on static prosthetics that neither replicate the tactile feel of human tissue nor simulate active bleeding. To enhance the realism of its training programs, BraveHeart First Aid has requested the development of a bleeding control simulator to replace existing plastic prosthetics and introduce more immersive training scenarios. The integration of this simulator will allow students to refine their bleeding intervention techniques in a controlled environment, thereby improving preparedness for real-life situations in which timely and effective action is critical to saving lives.

The Bleeding Control Simulator is designed to support three bleeding intervention techniques taught by the client. The first technique is compression-based bandaging, a non-intrusive intervention technique where the student tightly wraps a bandage around the site of bleeding. However, in severe cases this is often insufficient, so students may escalate their intervention to gauze-strip wound packing, an intrusive technique where gauze strips are forcefully inserted into the wound, forcing blood to soak the gauze strips before it can leave the body.

If wound packing is still insufficient, students can resort to applying a tourniquet, a life over limb technique where they tightly wrap a band above the wound site with sufficient pressure to cut off circulation to the affected limb. As shown in Figure 2, this prevents bleeding at the site of injury, though the lack of blood flow deprives the limb of oxygen, leading to necrosis that may necessitate amputation in severe cases.



Figure 2: Applying a tourniquet to an arm. Photo taken from [3].

The Bleeding Control Simulator circulates a fluid mimicking human blood through a procured medical prosthetic featuring perforations simulating wounds from which the fluid can exude. The simulator examines the placement, force of compression, and hold time of compression-based approaches while providing feedback to the student if their techniques are effective. If the bleeding intervention technique is insufficient, the simulator informs the student within seconds. Additionally, the simulator can replicate a spectrum of bleeding injuries, thereby challenging even medically experienced students to refine their technique beyond what would be expected in first aid curriculums.

BraveHeart First Aid intends to incorporate the Bleeding Control Simulator into their existing class schedules. Therefore, the device shall be compact, easily portable, and straightforward to clean following each training session. It must be compatible with a fluid medium that cost-effectively yet accurately replicates the visual and physical properties of human blood. These features will allow more students to apply their bleeding intervention techniques in realistic scenarios and develop the skills necessary to save the lives of those in urgent need.

Based on the client needs, training limitations, and physiological characteristics of bleeding injuries, a set of functional, safety, and economic requirements were developed to guide the design process. The project requirements were tabulated into fourteen distinct sections that outline parameters such as blood pressure and volumetric flow for enhanced realism, to quality-of-life features for effective integration of the simulator into first aid classes. These sections are listed below.

1. The simulator shall pump a fluid through a compact tubing system that flows into a medical training prosthetic of a forearm or leg intended to simulate blood loss up to 500 mL [12] through pre-designated bleeding site.
2. The simulator shall pump fluid to the bleeding site at a volumetric rate between 0 to 70 mL/min [18] of fake blood to simulate a range of bleeding scenarios.
3. The medical training prosthetic shall be made of a material sufficiently soft to quantifiably measure the blood loss and withstand a maximum pressure of 40 kPa [12] from the system using bandages for compressional treatment, and escalatory treatments including wound packing and tourniquet application.
4. The simulator shall monitor the total volume blood loss through the bleeding site during the simulation to evaluate the effectiveness of bleeding intervention techniques performed by students.
5. The simulator must produce a fluid flow ranging from 660 Pa to 18 kPa [18] at the bleeding sites to simulate blood pressures under normal to severe conditions in the limb recreated by the medical prosthetic.
6. The simulator must be manufacturable and deliverable to the location of BraveHeart First Aid courses for a maximum of \$1,850.
7. Fake blood shall be manufacturable at the site of first aid training using a bowl and mixing spoon for under 30 dollars per liter.
8. The simulator shall service all students in a first aid course within available time by ensuring a bleeding simulation lasts a maximum of 5 minutes per student.
9. Simulator must measure pressures applied to the medical prosthetic limb ranging between 660 Pa and 40 kPa [17].
10. The prosthetic limb shall comply with ISO 10993 for safe skin contact during training [15].
11. The simulator shall comply with IEC 60601-1 for all electronics equipment on board [7].
12. The simulator shall have audiovisual alarms for the following events:
 - 12.1. Pump failure.
 - 12.2. Applied pressures over 40 kPa during bleeding intervention techniques.
 - 12.3. Fluid flow pressures below 660 Pa.
13. The simulator shall offer a variety of injuries to simulate ranging in difficulty, where the difference in difficulty comes from increasing flow pressures and increased rates of blood loss. Easy injuries shall only require the use of compression-based bandaging techniques, while more complex injuries must justify escalation techniques to wound packing with gauze strips and tourniquet application.
14. The simulator shall be cleanable by flushing water through the system for no longer than 3 minutes.

Each of these requirements were tabulated and categorized, then quantitative targets and verification methods were identified to produce quantifiable results. This analysis is available in Table 1.

Table 1: Summarized project requirements, quantitative targets, and verification methods.

Category	Requirement	Quantitative Target / Range	Verification Method	Results (Y/N)
Function	System shall pump fluid through tubing into prosthetic limb to simulate bleeding	Up to 500 mL total blood loss	Functional testing	Y: 600 mL capacity
Performance	Adjustable bleeding flow rate	0–70 mL/min	Testing and Calibration	Y: 0-200 mL/min
Performance	Fluid pressure at bleeding site	660 Pa – 18 kPa	Pressure testing	Y: 6 kPa capacity
Performance	Measure applied pressure on prosthetic limb	660 Pa – 40 kPa	Sensor validation	Y: 40 kPa capacity
Functional	Monitor and record total blood loss during simulation	Continuous volume measurement	Functional testing	Y: Precise flow sensor
Function	Provide multiple injury difficulty levels	Difficulty increases via flow rate + pressure	Functional testing	Y: Multiple scenarios, though manual setup is required
Usability	Simulation duration per student	Less than 5 minutes	Time trial	Y: 3-minute maximum simulations
Usability	System shall have a cleaning procedure	Flush with water in $\leq$ 3 minutes	Cleaning test	Y: Run simulation with water
Materials	Prosthetic material must be soft and withstand treatment forces	Withstand 40 kPa compression	Mechanical testing	Y: Prosthetic includes rubber material
Cost	Total system manufacturing and delivery cost	Maximum budget of \$1,850	Cost analysis	Y: Costs below \$1,850
Cost	Fake blood must be manufacturable on-site	Maximum \$30/L	Cost analysis	Y: Cost of \$0.51/L
Safety	Prosthetic must be safe for skin contact	Comply with ISO 10993	Compliance review	Y: Prosthetic includes rubber material
Safety	Electronics must meet medical electrical safety	Comply with IEC 60601-1	Compliance review	Maximum voltage and current of 5V, 200 mA
Function	Provide audiovisual alarms for critical faults	Alarm triggers when pump fails, applied pressure > 40 kPa, flow pressure < 660 Pa	Functional testing	Includes buzzer, LEDs, on-screen prompts

2. Conceptual Designs

During the concept generation phase of the project, the Bleeding Control Simulator was broken down into four distinct subsystems as listed in Figure 3.

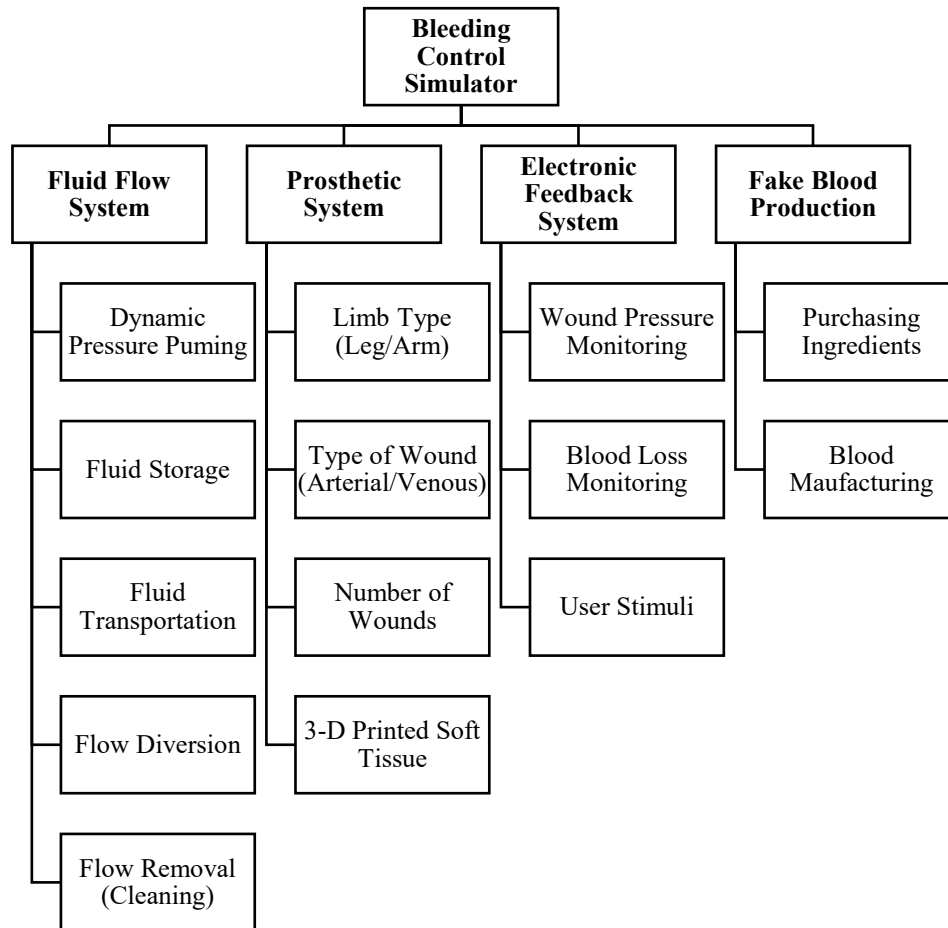


Figure 3: Breakdown of the Bleeding Control Simulator functions.

2.1. Fluid Flow System

One of the most pressing challenges for this project was designing a flow system capable of maintaining the necessary pressure and volumetric flow parameters regardless of whether the prosthetic outlet was blocked due to a successful bleeding intervention. The fluid flow system consisted of a reservoir, plastic tubing, and a pump to circulate simulated blood. From requirements two and three, the system was required to pump up to 70 mL/min at pressures up to 18 kPa. Three conceptual designs were produced that are outlined below to meet these requirements and constraints.

Design #1

- Uses a pump to circulate fake blood from a reservoir through tubing
- Flow is split at a Y-connector
- One path sends blood to the prosthetic bleeding site
- The other path returns blood to the tank for recirculation
- Design is simple and easy to assemble using purchased components
- However, it requires multiple external tubes
- Team wanted to reduce external tubing and overall component count

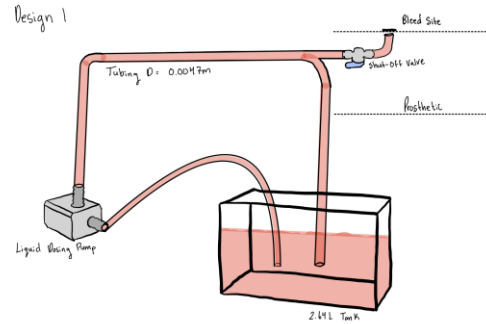


Figure 4: First fluid flow design (y-connector outside the prosthetic).

Design #2

- Y-connector moved inside the prosthetic for more realistic blood flow
- Returning tube would exit from inside the prosthetic
- Aims to better mimic the human circulatory system
- Requires modifying the prosthetic
- Modification is risky due to high cost of the prosthetic
- Increases assembly complexity despite improved realism

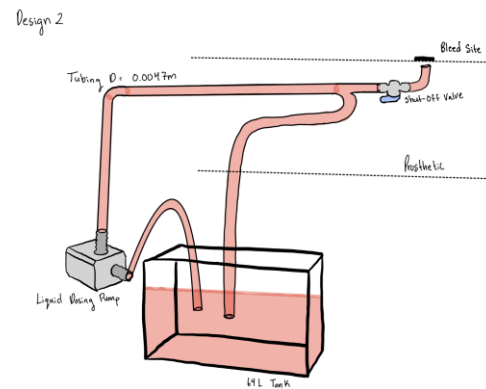


Figure 5: Second fluid flow design (y-connector inside the prosthetic).

Design #3

- Eliminates need for return flow to the tank
- Uses sensors to monitor flow conditions in real time
- Pressure increases when flow is blocked at the prosthetic exit
- Sensor detects pressure rise inside the tubing
- Microcontroller turns off pump until pressure drops
- Provides a more advanced and automated system
- Removes need for recirculation tubing
- Highly complex design to implement

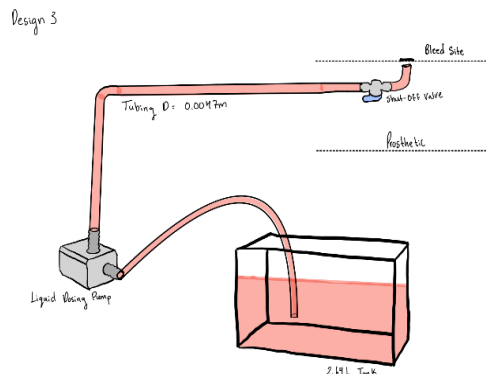


Figure 6: Third fluid flow design (eliminated y-connector).

The fluid flow system is integral to the Bleeding Control Simulator, as all other subsystems depend on its proper operation. Consequently, reliability was the primary concern when evaluating the available design options. The team expressed the greatest confidence in the first design due to its simplicity and ease of implementation, and this advantage outweighed the drawback of requiring an additional external tube outside the prosthetic. As a result, this design was selected for the final product.

2.2. Electronic Feedback System

For the electronic feedback system, the pressure applied to the wound and the volume of blood loss could be monitored using a pressure sensor and a water level sensor, respectively. These measurements enabled the system to provide feedback through LED indicators, audio signals, or on-screen displays. A summary of the different concepts is presented in Figure 7.

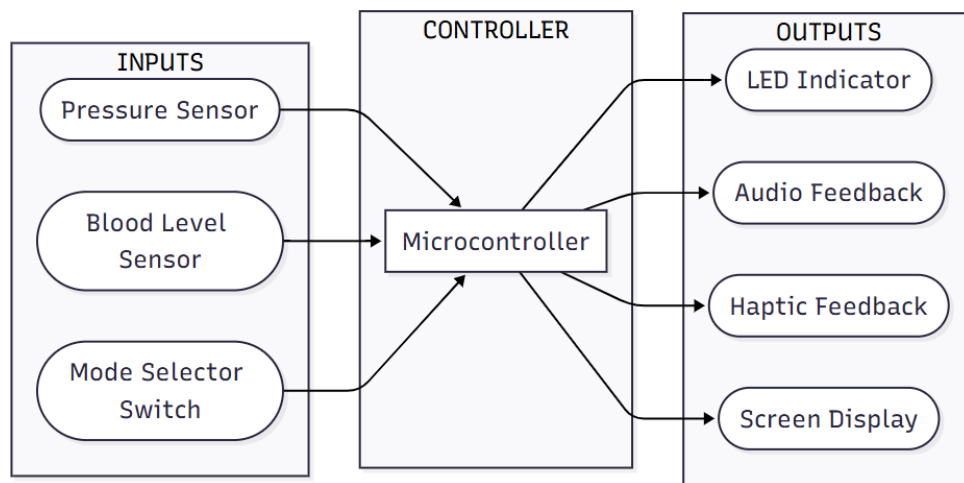


Figure 7: Conceptual inputs, controller, and outputs for the feedback system.

For the actual inputs the team initially ideated using the following concepts:

- Thin film sensors for the applied pressure by the user
- Flow sensor to measure how the flow changes if a tourniquet is applied

For the conceptual designs of the feedback system, each mechanism shown in Figure 7 presented distinct advantages and disadvantages for the project. A screen display offered the potential to provide the most comprehensive feedback through plots and on-screen prompts, but the team was concerned that an overly complex interface could divert the student's attention away from the bleeding injury.

In contrast, LED, haptic, and audio feedback could deliver immediate and easily interpretable signals but would limit the feedback structure to simpler on/off states. During the conceptual design phase, it was difficult to determine whether an optimal system would prioritize rapid, intuitive feedback at the expense of reduced nuance, or the opposite. Consequently, both approaches were developed in parallel and the preferred feedback structure was selected based on prototyping results and client feedback.

2.3. Fake Blood Production

A blood substitute for the Bleeding Control Simulator has three critical requirements:

- **Realism:** The substitute must closely replicate the visual appearance and viscosity of real blood.
- **Safety:** The substitute must not damage the fluid flow system during circulation or pose any risk to students in the event of spills.
- **Cost:** The substitute must be inexpensive to produce, as lost fluid must be replenished after each simulation.

Several existing recipes for simulated blood were researched, typically consisting of mixtures of water, thickening agents, and food colouring. A review of the literature identified multiple candidate formulations, as summarized in Table 2.

Table 2: Researched fake blood recipes.

Ingredient	Recipe 1	Recipe 2	Recipe 3
Powdered Sugar	454 g	—	—
Cocoa Powder	8g	—	—
Corn Syrup	—	750 mL	—
Coffee Creamer	—	890 mL	—
Vegetable Oil	—	—	50 mL
Food Colouring	1 oz	2 oz	1 oz
Water	800 mL	750 mL	900 mL

These recipes required prototyping to verify that they satisfied the three criteria outlined above. The primary concern regarding the blood substitute was viscosity, as the target formulation required a viscosity approximately four times that of water

2.4. Sensor Encapsulation Concept

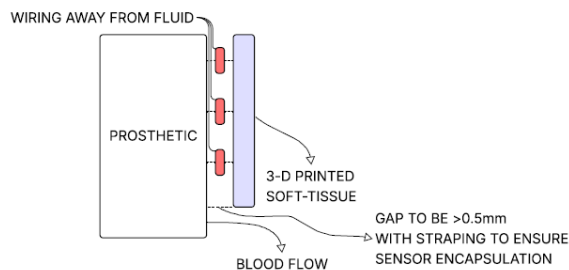


Figure 8: Sensor encapsulation schematic.

The 3-D printed soft-tissue component was developed as a protective and functional interface between the prosthetic structure and the embedded sensing elements. Its primary purpose is to provide a degree of waterproofing for the sensors while still allowing for accurate pressure transmission during simulated bleeding scenarios. This concept was to ensure that sensor readings remain reliable without being compromised by direct fluid exposure.

The initial design consisted of a concentric cylindrical geometry, featuring an inner cavity to house the prosthetic and an additional layer intended to encapsulate the sensors. The component was planned to be fabricated using thermoplastic polyurethane (TPU) selected for its flexibility and durability.

2.5. Initial Design Plan

This initial design schematic shown in Figure 9 represents the team's early conceptual approach to the system layout and component integration. At this stage, it serves primarily as a baseline framework to guide development rather than a finalized configuration. As the project progressed, the design evolved through iterative refinement, with changes driven by testing results, practical constraints, and performance requirements. All design decisions made during this process were clearly justified based on engineering principles, experimental data, and feasibility considerations.

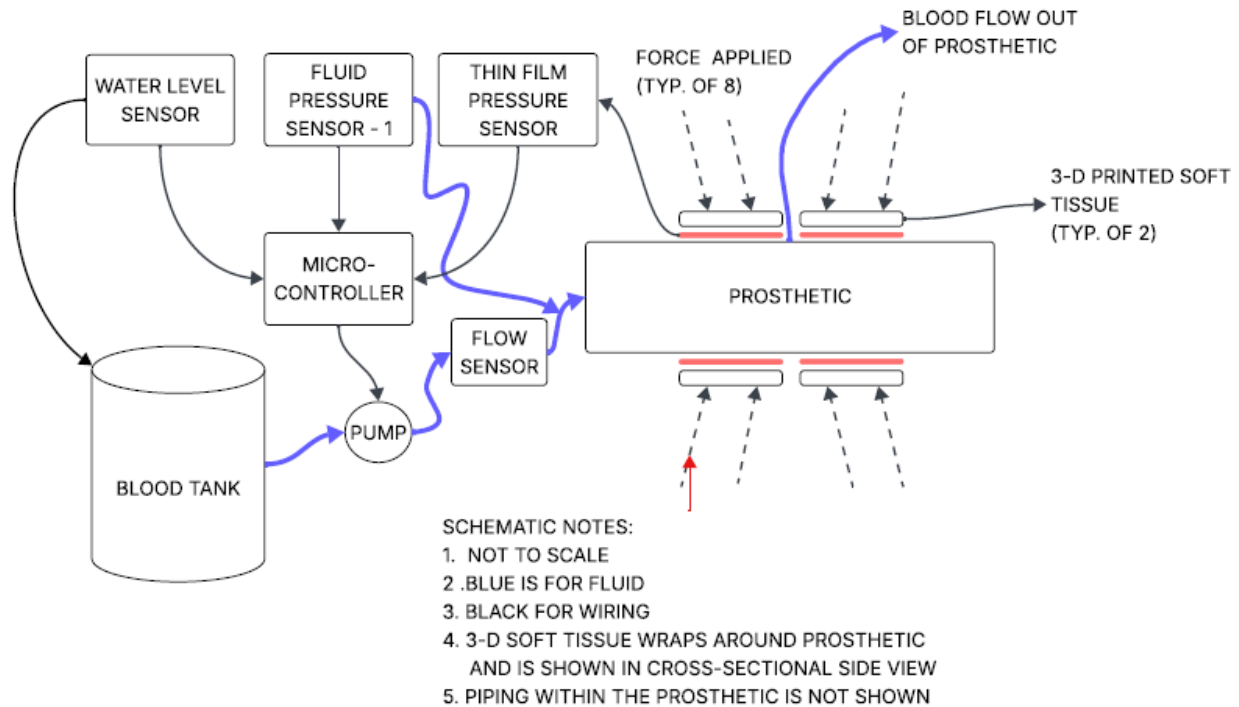


Figure 9: Initial design schematic.

Additionally, the integration of sensors shown in this schematic is preliminary. Each sensor went through a thorough verification and validation method to ensure accuracy, reliability, and suitability for the intended application. The methods and results of this verification process were documented and discussed in detail in later stages of the report.

3. Detailed Design and Iterations

Each of the subsystems outlined in the conceptual designs went through their own unique design process before arriving at their final designs. Some stages of the design process required in-depth prototyping, while other stages focused entirely to verification and validation to ensure the product met the requirements. A summary of this process is shown in Figure 10.

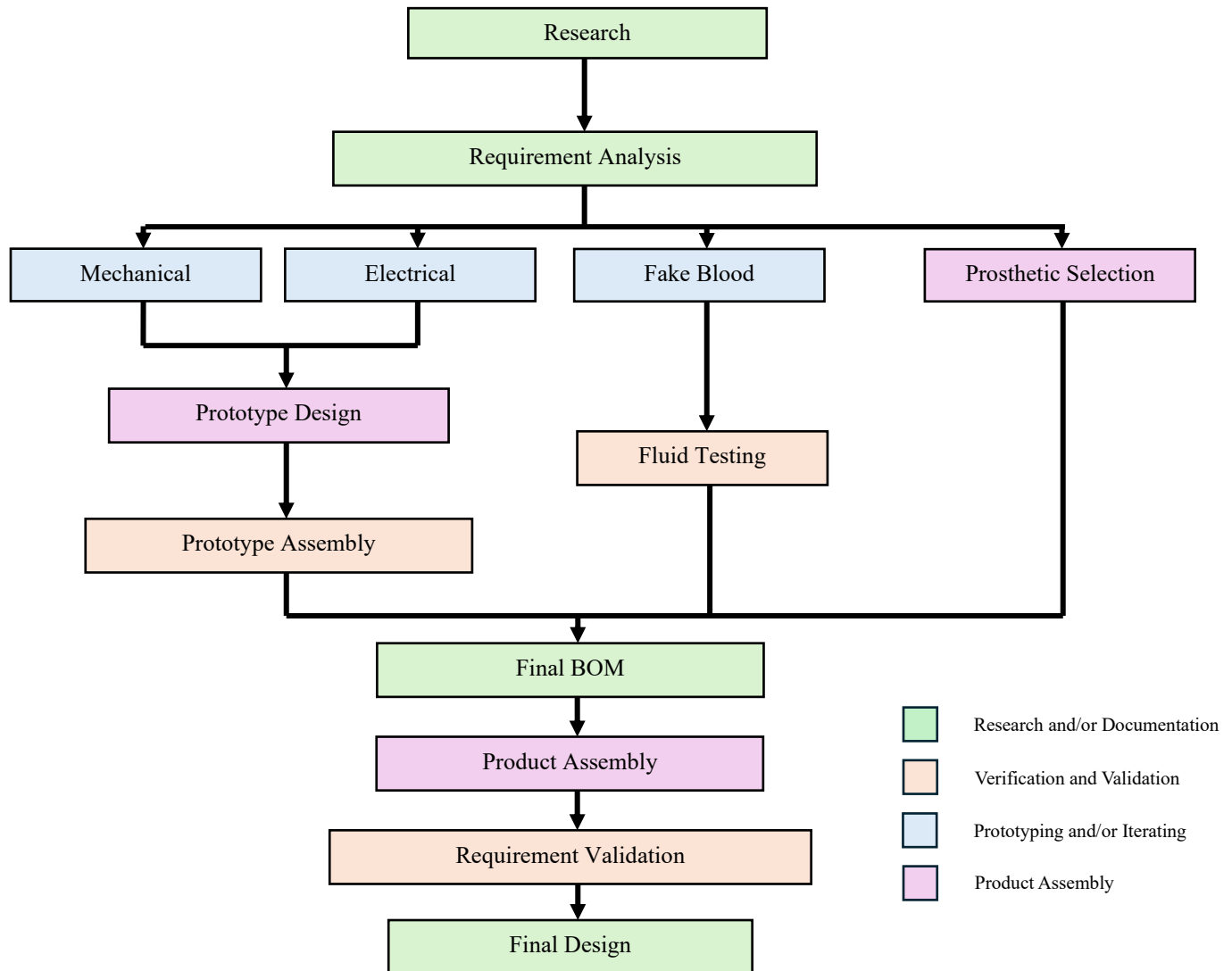


Figure 10: Design and construction process overview.

3.1. Mechanical - Pump and Tube Sizing for Arterial Flow

To determine the appropriate sizing of the pump and tubing required to circulate the simulated blood, the initial step was to analytically derive the tubing dimensions and pump capacity needed to achieve the target combination of volumetric flow rate and pressure.

For this project, the most demanding pump requirements were based on replicating the flow conditions of the ulnar artery, which experiences the highest pressures among the arteries of the arm due to its proximity to the heart. The pressure in the ulnar artery is approximately

120mmHg, or 16 kPa [4]. Therefore, the pump must be capable of delivering a flow rate sufficient to achieve this pressure. The tubing diameter is directly related to the resulting flow rate and pressure at the bleed site, meaning that the pump performance and tubing diameter must be selected in combination to satisfy the specified requirements. This section presents the calculations used to justify the selected pump

The average velocity of the fluid is solved using the Darcy-Weisbach equation for the pressure drop in a tube shown in Equation 1.

$$\Delta P = f \left(\frac{L}{D} \right) \left(\frac{\rho v^2}{2} \right) \quad (1)$$

where

- $\Delta P = 16$ kPa (pressure drop)
- f = Friction Factor (depends on Reynolds Number)
- $L = 1$ meter (length of tubing)
- $D = 0.0047$ meters (tubing diameter)
- $\rho = 1060$ Kg/m³ (blood density)
- v = Velocity of the fluid.

For this project, the friction factor in the tube is approximated using the Blasius resistance formula. However, the Reynolds Number must first be calculated using Equation 2.

$$Re = \frac{\rho v L}{\mu} \quad (2)$$

Following this the Blasius resistance formula yields the friction factor using Equation 3.

$$f = \frac{0.3164}{Re^{\frac{1}{4}}}. \quad (3)$$

The velocity value is an unknown in these equations thus is substituted back into the Darcy-Weisbach and yielded a value of:

$$v = 1.784 \text{ m/s}.$$

The flow rate is then solved using Equation 4.

$$Q = Av. \quad (4)$$

Using this for a circular tube, the flow rate equation yielded a flowrate of:

$$Q = 1857 \text{ mL/min}.$$

Although analytical calculations yielded a required flow rate of approximately 1857 mL/min to replicate arterial conditions, the final system was intentionally operated at lower flow rates to better align with training realism and safety constraints. The selected pump provides excess capacity, allowing adjustable flow conditions across multiple training scenarios while ensuring the system can meet peak performance requirements when necessary

3.2. Mechanical - Pump Selection

After completing these calculations, group members initially selected the Liquid Dosing Peristaltic Pump [1] for its advertised flow rates up to 6000 mL/min, as this would easily cross the 1857 mL/min threshold for the project. However, the main concern was that the pump was oversized, and that the high cost of \$369.86 was unjustifiable for the budget. Therefore, a smaller pump was used to prototype to determine if cheaper alternatives were viable.

The Pawfly submersible pump was used for early prototyping as shown in the top right of Figure 11. Preliminary testing demonstrated that the submersible pump could achieve flow rates up to 1536 mL/min, thereby eliminating the need for the more expensive peristaltic pump at this stage.

Despite producing sufficient flow for the Bleeding Control Simulator, the submersible pump was not suitable for the final design because it required connection to an electrical outlet. This limitation conflicted with the intended outdoor operation of the simulator, which was preferred to simplify cleanup from simulated blood loss. Also, introducing this wall power connection to the existing Arduino hardware setup would be electrically unsafe.

Therefore, the team iterated on the design to identify pump options that did not require direct access to a power outlet. The first alternative evaluated was a 3.7 V DC low-flow mini peristaltic pump, which was tested with the fully assembled prosthetic. Although this pump functioned as intended, its volumetric flow rate was insufficient for the project requirements.

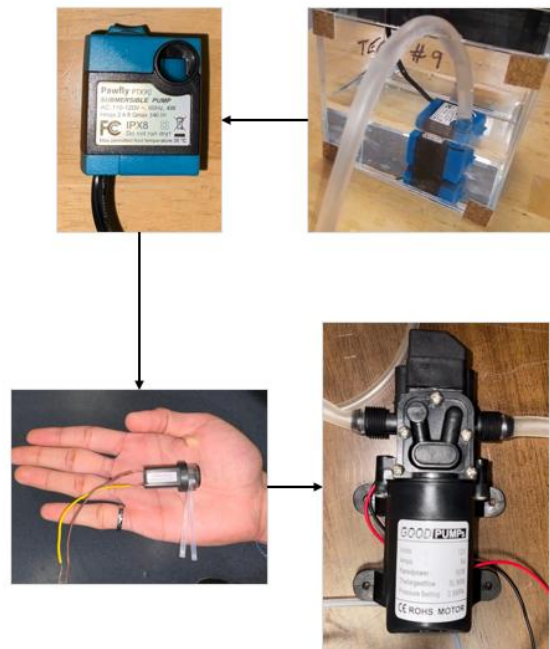


Figure 11: Pump iterations.

The final pump tested was a 12 V DC diaphragm pump shown on the bottom left of figure 11, which provided an adequate volumetric flow rate. A simple 12V DC power adapter along with a relay to turn the pump on or off is used. This configuration operates at 12 V DC and low current levels, ensuring safe operation and alignment with low-voltage electrical safety practices.

Table 3: Pump iteration process summarized.

Selected Pump (in chronological order)	Reason for Selection	Reason for Re-Iteration
Pawfly Submersible Pump	Low cost	Requires power outlet
Low Flow Rate Mini Peristaltic Pump 3.7V DC Motor	Low cost	Volumetric flow rate was far too low
DC 12V Diaphragm Pump	Can be controlled using Arduino, high volumetric flow	—

3.3. Mechanical - Sensor Encapsulation



Figure 12 : Sensor Encapsulation - Early Prototype

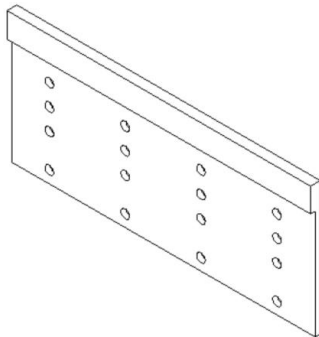


Figure 13 : Wrap - Around Strap for Encapsulation



Figure 14 : Final Sensor Encapsulation Strategy

Figure 12 shows an early prototype of 3-D Printed TPU to waterproof and electrically isolate the thin film sensors. Early testing revealed that the design did not achieve the desired level of softness and compliance required to realistically mimic human tissue. As a result, this component underwent a second iteration, with modifications to geometry, material properties, and fabrication parameters.

These modifications simply included making it a wrap-around sleeve around the prosthetic and also printing it thinner than before as shown in Figure 13. However, this continued to hinder the realism requirement of the Bleeding control simulator.

As a result, the final design shifted toward an external sensor placement strategy, where sensors were positioned around the wound region and secured using a saran wrap layer. This approach allowed the sensors to remain in close contact with the surface while being isolated from direct fluid exposure as shown in Figure 14. It also provided a practical and low-cost solution that avoided modifying or risking damage to the prosthetic itself.

At the same time, the use of an external wrapping layer effectively prevented water ingress into the sensor wiring, improving system reliability and ease of maintenance. The performance of this approach was validated through multiple pump tests, and through accurate sensor readings discussed later.

3.4. Electrical - Feedback System User Interface

The initial LED feedback system was proving to be inadequate, so an additional buzzer was added into the circuitry to make it more interactive, however adding more circuit elements into the already crowded breadboard added complexity for the client. So, an LED system but complemented with a MATLAB GUI to make it more classroom friendly as everyone could see on the screen what the instructor was doing was developed. This process is outlined in Figure 15



Figure 15: Feedback system iterations.

The feedback user interface was developed in MATLAB to monitor and visualize real-time sensor data from the prosthetic system. Five thin-film pressure sensors are embedded within the prosthetic, and their readings are first collected using an Arduino. These sensor values are then transmitted to the MATLAB environment for processing. In addition to the thin-film sensors, data from a pressure transducer and a volumetric flow sensor is acquired through the same pipeline.

The MATLAB user interface processes these incoming signals and generates real-time plots, allowing users to observe variations in pressure and flow throughout the system. This provides immediate visual feedback for performance analysis and system behavior.

During testing, the MATLAB interface functioned as a comprehensive monitoring and feedback tool that integrated multiple sensor inputs into a single, user-friendly visualization platform. However, combining the interface with the bleeding prosthetic created an overstimulating environment, as team members were often required to divide their attention between performing bleeding control techniques and monitoring multiple plots on the screen.

Based on client direction, and further testing the GUI environment was upgraded as shown in Figure 16, to include the following:

- A dynamic pressure bar that visually represents pressure magnitude
- A timer to track the duration of each simulation run
- Mode indicators showing whether the system is operating in dry, wet, or cleaning mode

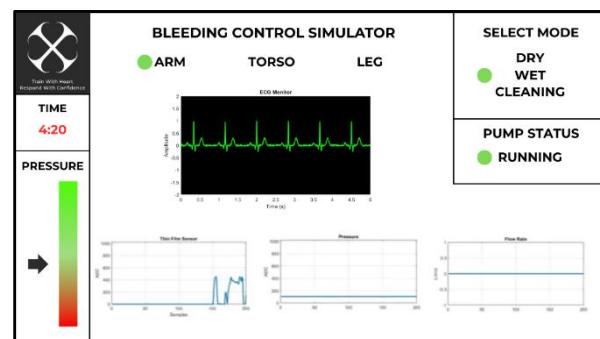


Figure 16: MATLAB user interface.

3.5. Prosthetic Selection

As directed by the client, the prosthetic was purchased from online retailers. One of the first decisions was to select either an arm or leg prosthetic. Considering that the major arteries in the arm have lower pressure and volumetric flow rates due to their position in the body relative to the heart [10], arm prosthetics were the preferred option because they require smaller pumps and tubing. This effectively reduced project costs without compromising the realism of the simulation.

After researching multiple prosthetics, three viable options were shortlisted as shown in Table 4. Selection criteria included cost, the size of the tubing entering the prosthetic, and the types of wounds simulated at the bleeding sites, as well as their alignment with scenarios taught in BraveHeart's curriculum. Although each prosthetic offered distinct advantages and disadvantages, the Translucent Haemorrhage Control Arm was determined to be the best overall fit and was therefore selected for the final product. The actual prosthetic arm is shown in Figure 17.

Table 4: Prosthetic comparison.

Prosthetic	Advantages	Disadvantages
Amputated L-Arm Sim 3G [7]	Low cost, lightweight	Small size which is difficult for tourniquet application. Lack of shoulder or hand reduces realism
Wound Packing Arm Trainer [2]	Variety of bleeding sites available, wearable strap for additional realism	Expensive. Wound sites are not designed after scenarios taught in BraveHeart first aid classes (gunshot wounds)
Translucent Haemorrhage Control Arm [8]	Low cost, heightened realism due to inclusion of elbow and hand, wound sites designed after scenarios taught in BraveHeart first aid classes (knife wounds)	White plastic material used to produce the prosthetic reduces realism



Figure 17: MedEduQuest prosthetic arm.

3.6. Final Design

The final design simulates realistic bleeding conditions and provides real-time feedback on the effectiveness of pressure application during hemorrhage control as required. A photo of the setup is available in Figure 18.

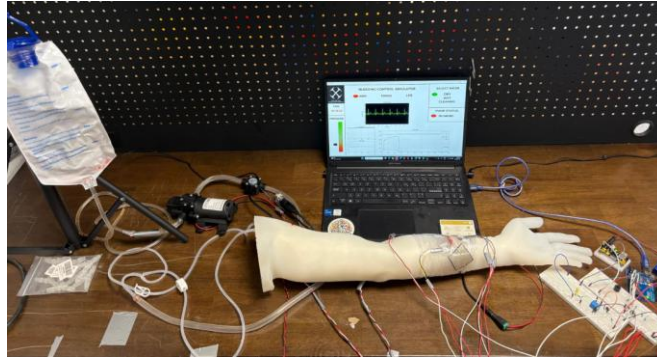


Figure 18 : Complete Final Design

Fluid Flow and Pump System

- The fluid delivery simulates blood flow using a diaphragm pump with a maximum capacity of 5 L/min.
- During normal operation, it achieves a target flow rate of 2000 mL/min, sufficient to mimic severe arterial bleeding for simulation purposes.
- The system uses a fluid reservoir/bag mounted on a stand, connected by tubing to the pump and subsequently to the simulated limb as can be seen in Figure 18.
- The flow is routed through the artificial limb as seen in Figure 16, allowing trainees to physically apply pressure and control simulated blood loss.

Sensor System

- The limb model includes five thin-film pressure sensors strategically placed around the simulated wound area as can be seen in Figure 19.
- These sensors measure the contact pressure applied by the trainee's hand or tourniquet.
- The data is transmitted to the computer interface, allowing accurate pressure range detection.
- The simulator software displays both the real-time pressure readings and the system status to guide the training process.

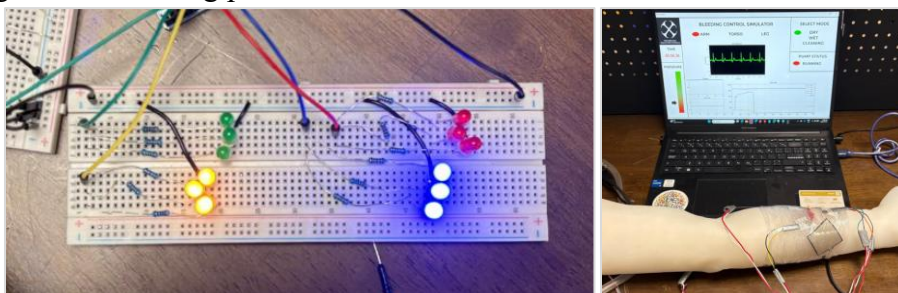


Figure 19 : Final Feedback System

Feedback and Control Interface

- The LED feedback circuit, built on a breadboard and interfaced with the sensor system, visually communicates the simulation's status:
- Red (blinking): Indicates active bleeding which means pressure has not yet reached the required threshold; immediate action needed.
- Green (solid): Represents correct pressure threshold has been applied.
- Yellow (blinking): Signals maintain pressure and the applied pressure is close to ideal but must be sustained for a specific duration.
- Green (blinking): Confirms requirement achieved which means correct pressure has been maintained long enough to stop the bleeding.
- Blue + Yellow (both solid): Marks simulation complete and the scenario ended successfully.
- This multi-step LED feedback system provides clear and intuitive guidance throughout the training., from recognizing active bleeding to successful hemorrhage control.

Integrated Operation

The pump initiates blood flow through the artificial wound.

- The sensors continuously measure applied pressure on the wound surface.
- Sensor data is transmitted to the control software, which determines the status (too low, correct, or stable).
- The LED system lights up correspondingly, providing instant feedback to the trainee.
- When the correct pressure is maintained for the required duration, the LEDs transition to blue and yellow, signaling a successful completion.

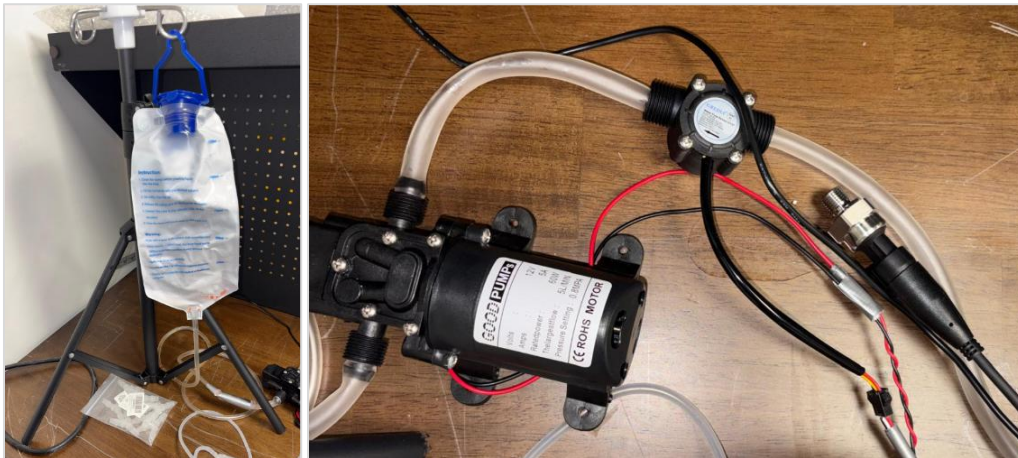


Figure 20 : Stand & surgical bag (left), pump, flow sensor and pressure transducer (right).

4. Verification and Testing

Group members ensured that the project met all the requirements using product tests throughout the project lifespan. This section outlines some of the tests and provided a summary table of how each requirement was verified.

4.1. Pump Testing

To ensure that the Bleeding Control Simulator can meet the volumetric flow requirement of at least 70 mL/min outlined in requirement two, a prototype was created using a submersible pump that moved water between two cooking pots on a counter. The setup is shown in Figure 21. The total volume of fluid moved into the second pot was recorded over thirty seconds to calculate the volumetric flow, and from this data, the volumetric flow was recorded as 1536 mL/min.



Figure 21 : Fluid flow system volumetric flow testing.

It is important to note that the volumetric flow in the final product was expected to be lower than the results recorded from this prototype because the fake blood would have a higher viscosity and travel a further distance before escaping through the prosthetic. Regardless, the recorded volumetric flow was over twenty times larger than the outlined requirement, so group members were confident that the final design would be sufficient even with those factors.

4.2. Sensor Calibration

Calibration of all sensors was performed using fundamental physical relationships to correlate raw sensor outputs with known physical quantities. Sources of experimental uncertainty include manual timing errors during measurements, sensor resolution limitations, and minor leakage within tubing connections. These factors are estimated to contribute an uncertainty of approximately $\pm 5\text{--}10\%$ in measured values which was reduced by conducting multiple trials.

4.2.1. Flow Sensor

The flow sensor was calibrated using a volumetric method, where a known volume of fluid was passed through the system and timed using a stopwatch. The flow rate was calculated as the change in volume over time. This procedure was repeated for multiple flow rates to establish a relationship between sensor output and actual flow rate as shown in Table 5.

Table 5 : Sensor calibration for the flow sensor.

Trial	Volume (L)	Time (s)	Flow Rate (L/s)	Sensor Output
1	1.0	10.2	0.098	48
2	1.0	8.5	0.118	58
3	1.0	6.7	0.149	72
4	1.0	5.5	0.182	89
5	1.0	4.8	0.208	101



$$Q = \frac{\Delta Volume}{\Delta Time}$$

Figure 22 : Flow sensor calibration method.

4.2.2. Pressure Transducer

The pressure transducer was calibrated using hydrostatic pressure principles. The sensor was placed at the base of a water tank, and the pressure was calculated using the known height of the water column and fluid density. Multiple water levels were tested to generate corresponding pressure values, allowing for the development of a calibration curve relating sensor output to pressure as shown in Table 6.

Table 6 : Sensor calibration for the pressure transducer.

Trial	Water Height (m)	Pressure (kPa)	Sensor Output
1	0.10	0.98	102
2	0.20	1.96	205
3	0.30	2.94	307
4	0.40	3.92	410
5	0.50	4.90	512



$$P = \rho gh$$

Figure 23: Pressure transducer calibration method.

4.2.3. Thin Film Sensors

Thin film pressure sensors were calibrated by applying known loads over their effective sensing areas. Using the manufacturer-provided sensor area, pressure was calculated as force divided by area. Known weights were applied incrementally, and the resulting sensor outputs were recorded. Multiple trials were conducted to ensure consistency and to generate calibration curves for each sensor.

Table 7 : Sensor calibration for thin film sensors

Trial	Mass (kg)	Force (N)	Pressure (kPa)	Sensor Output (ADC)
1	0.5	4.91	24.6	120
2	1.0	9.81	49.1	240
3	1.5	14.72	73.6	355
4	2.0	19.62	98.1	470
5	2.5	24.53	122.7	580

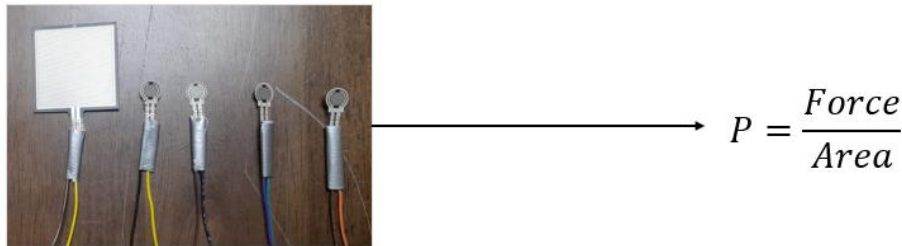


Figure 24: Thin film sensor calibration method.

Through these procedures, all sensors were verified to produce stable and repeatable outputs and were deemed suitable for integration into the system.

4.3. Sensor Encapsulation Verification

Encapsulation of the thin film sensors was achieved using a saran wrap barrier to protect against fluid exposure near the simulated bleeding site. Prior to testing, the baseline pressure introduced by the wrapping material onto the thin film sensors was accounted for during calibration.

Two validation tests were conducted to evaluate the effectiveness of the encapsulation:

- High-pressure exposure test: Sensors were subjected to direct water spray to simulate external fluid impact.
- Submersion test: Sensors were fully submerged in water to assess waterproofing performance under sustained exposure.

During testing, one sensor failed due to water ingress, indicating insufficient sealing. As a result, the encapsulation method was improved by reinforcing the wrapping with duct tape, which significantly enhanced durability and water resistance in subsequent trials.

4.4. Fake Blood Properties Testing

To verify the viscosity of the fake blood recipes, a portion of the prototyping budget was allocated to purchase ingredients for testing. The viscosity of the fake blood was approximated using the Efflux cup test [3], in which the time required for a fluid to drain through a funnel has an approximately linear relationship with viscosity. Since blood has a viscosity approximately four times greater than that of water [3], the ideal fake blood recipe would require four times longer than water to drain from the funnel.

To minimize the use of time and ingredients, an uncertainty of 25% was applied to the target drainage time for the fake blood. This resulted in a range of drainage times considered acceptable for achieving the desired viscosity. The derivation of these values is shown below.

$$\text{Water Drain Time} = 3.27 \text{ seconds}$$

$$\text{Fake Blood Drain Time} = \text{Water Drain Time} * 4 \pm 25\% = 13.08 \pm 3.27$$

$$\text{Acceptable Fake Blood Drain Times: } 9.81 \text{ to } 16.35 \text{ seconds}$$

Each of the samples was iteratively tested and the quantity of the recipe's thickening agent was altered until an acceptable drain time was achieved. The results are tabulated in Table 5.

Table 8: Efflux cup testing results.

Iteration	Recipe 1 Drain Time (s)	Recipe 2 Drain Time (s)	Recipe 3 Drain Time (s)
1	7.45	8.21	7.48
2	8.49	9.38	9.12
3	—	—	10.12
4	—	—	10.16

Early iterations revealed that the researched recipes produced viscosities that were too low to accurately replicate the viscosity of blood, necessitating the addition of a thickening agent beyond the original formulations. After the second iteration, it became clear that the first two recipes would require large quantities of thickening agent to meet the required threshold, which was not viable because the cost of these agents would exceed the maximum budget of thirty dollars per litre. In comparison, using vegetable oil as the thickening agent for the third recipe kept the cost manageable as additional oil was incorporated into the mixture. Consequently, the group discontinued work on the first two recipes and continued refining the third recipe until a batch achieved an acceptable drain time.

In addition to viscosity requirements, the fake blood needed to visually resemble human blood. While it is possible to quantitatively compare the visual similarity of fluids using colorimetry or spectrophotometry to assign numerical values to colour, these methods were considered excessive for this project, as visual resemblance is not integral to the function of the Bleeding Control Simulator. Furthermore, the appearance of blood varies between individuals depending on oxygen levels, nutritional intake, and related factors [11].

Therefore, a suitable fake blood recipe was selected in consultation with the client using photographs of the samples produced during the Efflux cup testing process. One of these samples is shown below in Figure 25.

Using these factors, the team determined the final recipe for the fake blood as shown in Table 6. This design is simplistic, inexpensive, and can be quickly assembled on-site at BraveHeart First Aid by adding the ingredients to a container and stirring.



Figure 24 : Final Fake Blood Recipe

4.5. Requirement Verification Summary Table

A table summarizing how the project met each of the designated requirements is available below.

Table 9: Summary of each requirement and the methods to verify quantifiable parameters.

Requirement Summary	Verification Method & Comments
Simulate blood loss up to 800 mL	Tank holds more than 500 mL. Simulations losing more than 500 mL were performed
Bleeding rates from the prosthetic up to 70 mL/min	Measured volumetric rates of fake blood from the bleeding site on the prosthetic
Prosthetic can withstand up to 40 kPa	Prosthetic made of durable plastic with a yield strength above the threshold
Track volume of blood lost through prosthetic	Installed a volumetric sensor in the tubing
Fluid flows between 660 Pa and 18 kPa	Installed a flow pressure sensor in the tubing
Simulator produceable for maximum of \$1850	Budget kept under \$1850
Fake blood cost under \$30/L	Cost of the fake blood per liter was calculated using prices from local grocery stores
Max 5 minutes per simulation	Simulation coded to end after 5 minutes
Measure applied pressures between 660 Pa and 40 kPa	Installed a pressure sensor for applied forces
Prosthetic complies with ISO 10993 for safe skin contact	The selected prosthetic is manufactured from materials commonly used in medical training devices and is expected to comply with ISO 10993 standards for safe skin contact.
Electronics compatible with IEC 60601-1	The system operates at ≤ 12 V DC and current levels, aligning with IEC 60601-1 safety considerations for electrical medical devices.
Audiovisual signals for important simulation events	Microcontroller programmed to respond to relevant events by lighting LEDs or displaying screen prompts
Simulator offers scenarios of varying difficulty	Tunable parameters in the simulation and alterable volumetric flow rates from the pump
Simulator cleanable with water in under 3 minutes	Cleaning mode programed in the microcontroller that turns off after 3 minutes

To ensure that the requirements can be achieved both in the testing laboratory and during the intended use, tests were repeated at the client's headquarters in Kentville. This assured that the product could successfully meet the requirements to both the group members and the client.



Figure 25 : Setting-up Simulator in Kentville

5. Engineering Economic Analysis

The simulator is costly to produce and requires ongoing funding to replace the simulated blood used during training. Consequently, the design must remain within the budgetary constraints of \$30/L for fake blood and \$1850 for the overall project. The ideal outcome leaves room in the budget to replace blood well into the future without requiring additional funding. Therefore, the financial circumstances for this project throughout its lifespan for the client is explored in this section.

5.1. Component Cost Breakdown

To ensure the project is leaving sufficient financial resources for the future, the cost of all the components purchased for the project was compiled below.

Table 10 : Cost breakdown of all the components.

Item	Cost
Arduino Starter Kit	71.99
Silicone Tubing	8.79
Ball Valve	16.15
Duct Tape	17.99
Glass Tank	29.65
Analog Pressure Transducer	20.79
Water Flow Sensor	18.99
Hot Glue Gun	11.69
Sensor Module Kit	31.99
12V DC Power Adapter	16.99
Peristaltic Pump	31.05
22 AWG Solid Core Copper Wire	22.99
Saran Wrap	5.99
DC 12V Diaphragm Pump	27.88
MedEduQuest Prosthetic Arm	945.70
Thin Film Sensors	26.50
Fittings	21.08
Total	\$1,326.21

The total manufacturing cost of the simulator was \$1,326.21, which is \$523.79 below the project budget limit of \$1,850.

5.2. Fake Blood Cost Analysis

As outlined in Section 4.2 the first two recipes were not chosen due to unacceptably high production costs. To calculate the cost of blood production for our client, the prices of relevant ingredients were sourced from local grocery stores. The prices for all the ingredients is listed in Table 11.

Table 11: Ingredients and associated costs for fake blood recipes.

Ingredient	Quantity	Cost
Powdered Sugar	1000 g	\$4.56
Cocoa Powder	227 g	\$5.70
Corn Syrup	473 mL	\$3.42
Coffee Creamer	890 mL	\$14.79
Vegetable Oil	5L	\$14.80
Food Colouring	1 oz	\$4.75

Using these figures, the calculated price for each recipe is shown in Table 12.

Table 12: Costs to produce each blood recipe per liter basis.

Ingredient	Recipe 1	Recipe 2	Recipe 3 (modified)
Powdered Sugar	454 g	—	—
Cocoa Powder	8g	—	—
Corn Syrup	—	750 mL	—
Coffee Creamer	—	890 mL	—
Vegetable Oil	—	—	150 mL
Food Colouring	1 oz	2 oz	0.135 oz
Water	800 mL	750 mL	900 mL
Cost	\$30.72/L	\$26.02/L	\$0.51/L

Considering that the third recipe costs less than ten percent of the other recipes, it was the clear choice from a financial perspective. Using either of the first two recipes would exhaust the \$50 budget allocated for fake blood supplies after only a few simulations, causing the Bleeding Control Simulator to quickly incur expenses beyond acceptable limits.

To benchmark the in-house blood recipe against commercially available alternatives, a spot check of simulated products marketed for training use was conducted. Several brands of artificial blood are available, primarily in the United States, including Gaumard Artificial Blood Concentrate [5], VATA Simulated Blood [15], and NASCO Simulation Blood [9]. Although these alternatives would satisfy the realism criteria in both appearance and viscosity, their high costs would drive the project over budget and complicate the procurement process due to the need for international ordering.

Table 13: Costs of commercially available blood substitutes.

Brand	Cost (CAD)
Gaumard [5]	\$7.72/L
VATA Option 1 [16]	\$9.81/L
VATA Option 2 [17]	\$12.65/L
NASCO [9]	\$18.01/L

Compared with the project-selected formulation at \$0.51 per litre, Table 12 indicates that the in-house recipe is approximately 91% to 96% less expensive on a per-litre basis, supporting its suitability for repeated classroom use.

5.3. Consumables Budget Projection

Despite the lower cost of the third recipe, BraveHeart will inevitably need to invest additional funds in ingredients to replace the blood lost during training. To demonstrate the simulator's affordability, an estimate was developed to predict how long the \$50 allocation would last. This estimate is based on the following assumptions:

- Ingredient costs will remain constant
- Water has a negligible cost
- Fake blood is not recycled after use in the Bleeding Control Simulator
- Each simulation loses an average of 500 mL of fake blood
- Classes of first aid students comprise an average of ten students who each perform one simulation

Table 14: Calculations for replacing lost blood with a \$50 budget.

Parameter	Value
Fake Blood Cost	\$0.51/L
Total Blood Supply	98L
Number of Performable Simulations	180 Simulations
Number of Classes with Simulations	18 Classes
Estimated Cost per Student Simulation	~\$0.26 per student simulation

Considering that BraveHeart typically runs one or two first aid classes on weekends, the ability of the \$50 budget to support eighteen classes indicates that sufficient funds are available to produce simulated blood for several months before additional ingredients must be purchased. This demonstrates that the project will remain affordable for the foreseeable future despite ongoing operating expenses.

6. Handoff Plan & Final Design Technical Documentation

The final design consists of an integrated hydraulic simulation system, sensor network, data acquisition unit, and a real-time feedback interface developed in MATLAB. The system is designed to simulate hemorrhage scenarios and provide users with immediate feedback on their ability to apply appropriate pressure to stop bleeding.

The design enables users to practice hemorrhage control techniques by applying pressure to a prosthetic model. System performance is evaluated based on whether the applied pressure exceeds a predefined threshold and is sustained for a sufficient duration. Additionally, the system incorporates flow and pressure monitoring to evaluate the effectiveness of interventions such as tourniquet application.

6.1. Hydraulic System Design

The hydraulic subsystem simulates blood flow through a controlled piping network. Fluid is driven by a pump through tubing that represents a bleeding vessel within the prosthetic. A pressure transducer is installed inline to measure static pressure within the system, while a volumetric flow sensor measures the flow rate of the fluid. These measurements allow the system to determine whether bleeding has been successfully reduced or stopped.

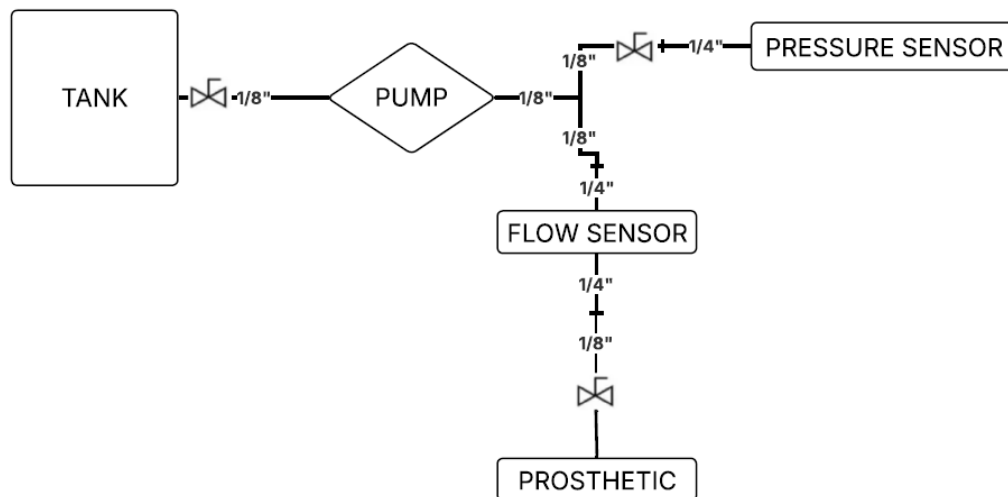


Figure 26: Hydraulic system with specified tube diameters and locations of electronic sensors.

By monitoring both pressure and flow, the system provides a realistic representation of hemorrhage control, where effective intervention results in reduced flow and increased internal resistance.

6.2. Data Acquisition and Hardware Architecture

Sensor data is acquired and processed using an Arduino Uno R3, which serves as the central data acquisition unit.

Sensors Used:

- Five thin-film pressure sensors: Placed around the bleeding site to measure externally applied pressure
- Pressure transducer (0–80 PSI): Measures internal system pressure
- Flow sensor (1–30 L/min): Measures volumetric flow rate

Data Acquisition Process:

1. Sensors output analog voltage signals
2. The Arduino performs analog-to-digital conversion
3. Data is transmitted via serial communication (USB, 115200 baud)
4. MATLAB receives and processes the data

This architecture enables continuous and near real-time data transfer between hardware and software components.

6.3. Code Flow and Program Logic

The software implementation follows a structured data processing pipeline to ensure accurate and responsive system behavior. This logical structure ensures that the system operates continuously and provides immediate feedback to the user.

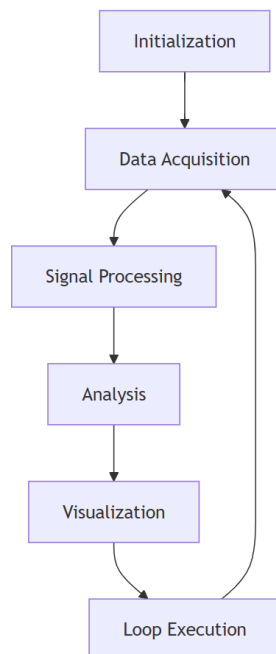


Figure 27: Flowchart for the electronic feedback system.

6.4. Feedback User Interface

The graphical user interface (GUI) developed in MATLAB serves as the primary interaction point between the user and the system.

The GUI includes:

Real-time plots of:

- Pressure from five thin-film sensors
- Flow rate
- Internal pressure
- A dynamic pressure bar indicating applied force
- A timer tracking duration of pressure application
- Mode indicators (dry, wet, cleaning)
- Display of injury location

A pressure threshold is clearly shown on the plots, allowing users to visually assess whether sufficient pressure is being applied.

6.5. Functionality

The interface updates continuously as new data is received, enabling real-time visualization of system behavior. This allows users to immediately adjust their actions based on feedback, reinforcing proper technique.

6.6. System Integration

The final system integrates hydraulic, electronic, and software components into a unified feedback loop:

- The hydraulic system generates measurable flow conditions
- Sensors capture both internal and external system responses
- The Arduino Uno R3 processes and transmits data
- MATLAB visualizes the data in real time

This closed-loop interaction allows users to directly observe the impact of their actions on system behavior, making the design effective for training applications.

7. Future Considerations

The final design meets the primary objectives established for the Bleeding Control Simulator and provides BraveHeart First Aid with a portable, functional, and realistic training device. However, as with any prototype developed within limited time and budget constraints, there remain several opportunities for refinement in future iterations of the project. These improvements would primarily focus on increasing physiological realism, expanding the range of measurable student performance metrics, and further improving the simulator's ease of use in repeated classroom settings. Since the current design already demonstrates the feasibility of combining a bleeding prosthetic, fluid flow system, and feedback electronics into a single training platform, future work can build directly on this foundation to create a more advanced and polished product.

7.1. Pulsating Flow of Fake Blood

The interim design report outlined ambitions for the fluid flow system to produce a pulsating flow to mimic the expansions and contractions of a beating heart. This system selected systolic and diastolic pressures and aimed to recreate them using the following formula

$$P(t, T) = P_{diastolic} + (P_{systolic} - P_{diastolic}) \frac{1 - \cos\left(\frac{2\pi t}{T}\right)}{2} \quad (6)$$

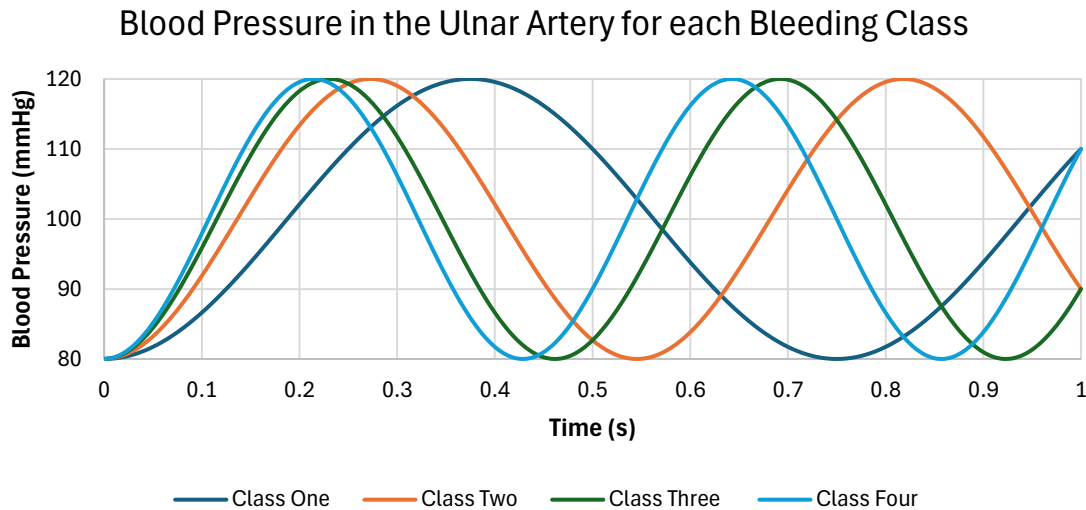


Figure 29: Mathematical re-creation of pulsating flow for various classifications of blood loss.

Where t was measured in seconds and T was an integer that decreased as blood was lost from the system, mimicking how the circulatory system increases its heart rate to maintain consistent blood pressures when losing high quantities of blood.

This feature could not be implemented into the final design due to the increasing list of requirements of the pump. It was outlined in earlier sections how to pump needed to be inexpensive, deliver approximately 1857 mL/min of fake blood, and operate without a power outlet.

Adding the additional requirement of producing quickly alternating flow pressures was infeasible as not all these requirements could be met simultaneously. Pumps that could deliver the pulsating flow at sufficiently high volumetric flows were available, though the costs made it impractical for the project. The group had decided that one of these features had to be eliminated, and the pulsating flow was chosen due to its non-necessity for the project. In future iterations for the project, this feature could be re-introduced using a pulsating pump, or by introducing shut-off valves that reduce the available area in the pipes for flow. However, there was insufficient time and funds for these options to be considered within the scope of the project.

7.2 Portability

One large future consideration is the portability of the simulator. As of the final design, the simulator must be plugged into a wall outlet and an extension cord must be used to move the simulator away from the wall as the pump adapter cord is approximately 5 feet. To allow the simulation to take place without a need for a wall plug, a battery system would need to be implemented. This could be as simple as an AC portable power bank incorporated into the simulator that could be charged before training sessions. However, the portability aspect would add weight to the overall simulator and increase the budget. The trade-off is being able to use the simulator in any location.

7.3. Software

To make the simulator more compatible a software change from the current MATLAB GUI to Python should be integrated. After discussing with clients, it was found that they do not currently have a license for MATLAB, therefore, to use the GUI they will need to purchase a license. It was also noted that python is used more regularly in industry use, thus, a python integrated GUI would make the simulator much more compatible and user friendly with future client's needs.

7.4. Wire Management

The final simulator design has many wires from the sensors connected to the simulator. Upon testing at the Braveheart facility, it was noted that the number of exposed wires connected to the bleed site make the bleeding intervention techniques such as tourniquets and wound packing much more difficult. To ultimately eliminate this problem moving forward, the wires and sensors should be implemented into the prosthetic and under the "skin". The "skin" would include a sleeve design placed over the prosthetic and to act as wire management. This would ensure that all wiring would not be directly exposed and in the path of the trainee, thus, eliminating the risk of accidentally effecting the wiring.

8. Final Design Technical Documentation and Project Handoff Plan

The formal hand-off of the Bleeding Control Simulator to BraveHeart First Aid is scheduled for April 9th, 2026. At the conclusion of the project, the simulator will be delivered as a fully assembled and functional prototype, accompanied by a complete and structured documentation package to support operation, maintenance, and future development.

8.1 Handoff Deliverables

The final deliverables will include both physical components and digital documentation, verified through a structured handoff checklist at the time of transfer.

Physical System Components:

- Prosthetic arm with integrated wound interface
- Fluid circulation system, including pump and tubing network
- Fluid reservoir and containment system
- Pressure and volumetric sensing components
- Feedback interface electronics (LED indicators and control hardware)
- Fully assembled and wired system

Digital and Documentation Package:

- Final technical report
- Engineering drawings of sensor encapsulation strategies as shown in section 3.3
- Bill of Materials as shown in section 5
- Arduino and MATLAB source code. The complete control logic and data processing scripts are provided in the Appendix section.
- Fake blood formulation and preparation procedure as shown in section 2 and 4.

A user operation manual, cleaning and maintenance guide, and troubleshooting and calibration guide have been developed and are included as part of the final documentation package.

All digital files will be organized into a clearly labeled folder structure labelled as Code, Documentation, Testing. These files will be transferred electronically to ensure accessibility and ease of future modification.

8.2 Handoff Procedure

To ensure a smooth transition, the following structured handoff process will be conducted:

Delivery of the fully assembled simulator and verification of all physical components using a checklist and the transfer of all digital files and documentation to the client

Live demonstration of system setup, operation, and shutdown which was already completed in the March 27th Kentville Visit. This will be followed by a guided walkthrough allowing BraveHeart staff to operate the system independently

8.3 Safety Protocols

To ensure safe and reliable operation, the following safety considerations and procedures will be communicated to the client:

- Identified Risks:
 - Presence of fluid near electrical components
 - Potential for leakage or tubing disconnection
 - Slip hazards due to spilled fluid
- Mitigation Measures:
 - Use of low-voltage electrical systems
 - Secure tubing connections and sealed interfaces
 - Stable system enclosure to prevent tipping
- Emergency Procedures:
 - Immediate system shutdown via power disconnection
 - Isolation of fluid reservoir in case of leakage
 - Cleanup protocol for spilled fake blood

These safety guidelines will be included in the user manual and reinforced during the handoff demonstration.

8.4 Cleaning, Maintenance, and Consumables

To maintain system performance and hygiene, the following procedures will be implemented:

- After each training session, the system must be flushed using the designated cleaning mode to remove residual fake blood from tubing and internal pathways
- Periodic inspection of tubing connections, pump functionality, and sensor integrity is recommended
- Sensors should be recalibrated as needed using the provided calibration procedures

Consumable materials, which only include fake blood ingredients can be replenished using the specifications provided in this report.

8.5 Testing and Validation Transfer

The client will be provided with a summary of system testing and validation performed during development. This includes:

- Verified flow rate ranges achievable by the pump system
- Sensor calibration relationships and expected accuracy ranges
- Functional validation of the feedback system

In addition, procedures for re-testing and recalibrating the system will be included. The handoff plan is designed to ensure that BraveHeart First Aid is equipped not only with a completed prototype, but also with the technical knowledge required for long-term use and future improvements.

9. References

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10. Appendix

Final developed version of Arduino Code for Bleeding Control Simulator

```
// ----- PIN DEFINITIONS -----
// LEDs
const int greenPin = 2;
const int yellowPin = 4;
const int redPin = 7;
const int bluePin = 8;
// Pump / Motor
const int motorPin = 9;
// Sensors
const int pressureSensorPin = A5;
const int thinFilmSensorPin0 = A0;
const int thinFilmSensorPin1 = A1;
const int thinFilmSensorPin2 = A2;
const int thinFilmSensorPin3 = A3;
const int thinFilmSensorPin4 = A4;
const int flowSensorPin = 3;
// ----- FLOW SENSOR VARIABLES -----
volatile int pulseCount = 0;
volatile int pumpState = 1;
unsigned long previousFlowTime = 0;
const unsigned long flowInterval = 250; // ms
float calibrationFactor = 450.0;
// ----- STATE + TIMING -----
int state = 0; // 0 = bleeding, 1 = threshold reached, 2 = maintained
unsigned long thresholdStartTime = 0;
unsigned long lastBlinkTime = 0;
const unsigned long blinkInterval = 500;
bool redState = false;
bool yellowState = false;
bool greenState = false;
unsigned long startTime;
const unsigned long runDuration = 1UL * 60UL * 1000UL; // 1 minute
// ----- INTERRUPT FUNCTION -----
void countPulse() {
    pulseCount++;
}
// ----- SETUP -----
void setup() {
    pinMode(greenPin, OUTPUT);
    pinMode(yellowPin, OUTPUT);
    pinMode(redPin, OUTPUT);
    pinMode(bluePin, OUTPUT);
```

```

pinMode(motorPin, OUTPUT);
pinMode(flowSensorPin, INPUT);
attachInterrupt(digitalPinToInterrupt(flowSensorPin), countPulse, RISING);
Serial.begin(115200);
Serial.println("Pressure,FlowRate,TF0,TF1,TF2,TF3,TF4");
startTime = millis();
}
// ----- LOOP -----
void loop() {
  unsigned long currentTime = millis();
  // ----- STOP AFTER TIME -----
  if (currentTime - startTime >= runDuration) {
    digitalWrite(redPin, LOW);
    digitalWrite(yellowPin, HIGH);
    digitalWrite(greenPin, LOW);
    digitalWrite(bluePin, HIGH);
    digitalWrite(motorPin, LOW);
    while(true);
  }
  // ----- READ EVERY 250 ms -----
  if (currentTime - previousFlowTime >= flowInterval) {
    // ----- FLOW -----
    noInterrupts();
    int pulses = pulseCount;
    pulseCount = 0;
    interrupts();
    float flowRate = (pulses / calibrationFactor) * (1000.0 / flowInterval);
    float flowRateLMin = flowRate * 60.0;
    // ----- SENSORS -----
    int pressureReading = analogRead(pressureSensorPin);
    int tf0 = analogRead(thinFilmSensorPin0);
    int tf1 = analogRead(thinFilmSensorPin1);
    int tf2 = analogRead(thinFilmSensorPin2);
    int tf3 = analogRead(thinFilmSensorPin3);
    int tf4 = analogRead(thinFilmSensorPin4);
    // ----- SERIAL -----
    Serial.print(pressureReading);
    Serial.print(",");
    Serial.print(flowRateLMin);
    Serial.print(",");
    Serial.print(tf0 - 500);
    Serial.print(",");
    Serial.print(tf1);
    Serial.print(",");
    Serial.print(tf2);

```

```

        Serial.print(",");
        Serial.print(tf3);
        Serial.print(",");
        Serial.print(tf4);
        Serial.print(",");
        Serial.println(pumpState);
    // ----- AVERAGE -----
    float avgTF = (tf0 + tf1 + tf2 + tf3 + tf4) / 5.0;
    // ----- BLINK TIMER -----
    bool blinkTrigger = false;
    if (currentTime - lastBlinkTime >= blinkInterval) {
        lastBlinkTime = currentTime;
        blinkTrigger = true;
    }
    // ===== FIXED STATE MACHINE =====
    // ----- STATE 0: BLEEDING -----
    if (state == 0) {
        // Red blinking
        if (blinkTrigger) {
            redState = !redState;
            digitalWrite(redPin, redState);
        }
        digitalWrite(yellowPin, LOW);
        digitalWrite(greenPin, LOW);
        digitalWrite(motorPin, HIGH);
        // If threshold reached → START timer and go to state 1
        if (avgTF > 300) {
            state = 1;
            thresholdStartTime = currentTime;
        }
    }
    // ----- STATE 1: MUST MAINTAIN PRESSURE -----
    else if (state == 1) {
        // If pressure drops → RESET to state 0 immediately
        if (avgTF <= 200) {
            state = 0;
            return; // exit early to avoid executing rest
        }
        // Green ON
        digitalWrite(greenPin, HIGH);
        digitalWrite(redPin, LOW);
        // Yellow blinking
        if (blinkTrigger) {
            yellowState = !yellowState;
            digitalWrite(yellowPin, yellowState);
        }
    }

```

```

    }
    // Check if maintained for 40 sec
    if (currentTime - thresholdStartTime >= 10000) {
        state = 2;
    }
}
// ----- STATE 2: SUCCESS -----
else if (state == 2) {
    digitalWrite(redPin, LOW);
    digitalWrite(yellowPin, LOW);
    digitalWrite(motorPin, LOW);
    pumpState = 0;
    // Green blinking
    if (blinkTrigger) {
        greenState = !greenState;
        digitalWrite(greenPin, greenState);
    }
}
previousFlowTime = currentTime;
}
}

```

Final developed version of MATLAB Code for Bleeding Control Simulator

```

clc; clear; close all;
%% ----- MODE SELECTION -----
mode = input('Select mode [DRY(1), WET(2), CLEANING(3)]: ');
% Default position
circle2_pos = [0.78 0.756];
if mode == 1
    circle2_pos = [0.78 0.85]; % DRY
elseif mode == 2
    circle2_pos = [0.78 0.8]; % WET
elseif mode == 3
    circle2_pos = [0.78 0.756]; % CLEANING
end
%% ----- FIGURE + BACKGROUND -----
fig = figure('Name','Simulator','NumberTitle','off',...
    'MenuBar','none','ToolBar','none',...
    'WindowState','maximized');
bg = imread('1.png');
ax_bg = axes('Position',[0 0 1 1]);
imshow(bg,'Parent',ax_bg);

hold(ax_bg,'on')

%% ----- TIMESTAMP -----
stopwatchText = text(ax_bg, 0.067, 0.65, '00:00:00', ...
    'Units','normalized','HorizontalAlignment','center',...
    'VerticalAlignment','middle','Color','black',...
    'FontSize',24,'FontWeight','bold',...
    'BackgroundColor','white','Margin',5);
%% ----- ARROW -----
arrowImg = imread('arrow.png');
arrow_x = 0.05;
arrow_y = 0.48;
arrow_w = 0.08;
arrow_h = 0.08;
arrow_ax = axes('Position',[arrow_x-arrow_w/2, arrow_y-arrow_h/2, arrow_w, arrow_h]);
imshow(arrowImg,'Parent',arrow_ax);
axis(arrow_ax,'off')
%% ----- CIRCLES -----
circle_radius = 0.02;
circle_pos = [
    0.21 0.84; % Circle 1
    circle2_pos; % Circle 2 (mode dependent)
    0.78 0.565 % Circle 3 (pump/status)
];
circles = gobjects(1,3);
for i = 1:3
    circles(i) = annotation('ellipse',...
        [circle_pos(i,1)-circle_radius, circle_pos(i,2)-circle_radius,...
        2*circle_radius, 2*circle_radius],...
        'FaceColor',[1 0 0],... % default red
        'EdgeColor','none');
end
% Set Circle 2 GREEN based on mode

```

```

set(circles(2),'FaceColor',[0 1 0]);
%% ----- ARDUINO SETUP -----
disp('Press ENTER to start')
pause
s = serialport("COM8",115200);
configureTerminator(s,"LF");
pause(2)
disp("Arduino Connected")
%% ----- DATA STORAGE -----
windowSize = 200;
pressure = nan(1,windowSize);
flow = nan(1,windowSize);
thinfilm = zeros(5, windowSize);
t = 1:windowSize;
%% ----- PLOTS -----
ax1 = axes('Position',[0.15 0.26 0.2 0.15]);
p1 = plot(ax1,t,pressure,'LineWidth',2);
title(ax1,"Pressure"); grid(ax1,"on"); ylim(ax1,[0 1023]);
ax2 = axes('Position',[0.15 0.065 0.2 0.15]);
p2 = plot(ax2,t,flow,'LineWidth',2);
title(ax2,"Flow Rate"); grid(ax2,"on");
ax3 = axes('Position',[0.4 0.065 0.5 0.3]);
p3 = plot(ax3, t, thinfilm, 'LineWidth',2);
legend(ax3, ["TF1", "TF2", "TF3", "TF4", "TF5"])
title(ax3,"Thin Film"); grid(ax3,"on"); ylim(ax3,[0 1023]);
set(p3, {'YData'}, num2cell(thinfilm,2))
yline(ax3, 200, '--r', 'Threshold')
%% ----- MAIN LOOP -----
startTime = tic;
while true
    elapsed = toc(startTime);
    mins = floor(elapsed/60);
    secs = floor(mod(elapsed,60));
    millis = floor((elapsed - floor(elapsed))*100);
    timeStr = sprintf('%02d:%02d.%02d', mins, secs, millis);
    set(stopwatchText,'String',timeStr);
    % Stopwatch color
    if elapsed < 10
        set(stopwatchText,'Color',[0 1 0])
    elseif elapsed < 20
        set(stopwatchText,'Color',[1 1 0])
    else
        set(stopwatchText,'Color',[1 0 0])
    end
    line = readline(s);
    data = str2double(split(line,""));
    % NOW EXPECTING 8 VALUES
    if length(data)==8 && all(~isnan(data))
        pressure = [pressure(2:end) data(1)];
        flow      = [flow(2:end) data(2)];
        newThin = reshape(data(3:7),5,1);
        thinfilm = [thinfilm(:,2:end), newThin];
        statusFlag = data(8); % NEW VARIABLE
        set(p1,'YData',pressure)
        set(p2,'YData',flow)
    end
end

```

```

set(p3, {'YData'}, num2cell(thinfil, 2));
%% ----- CIRCLE LOGIC -----
% Circle 1 (thin film example)
if thinfil(end) > 200
    set(circles(1), 'FaceColor', [0 1 0])
else
    set(circles(1), 'FaceColor', [1 0 0])
end
% Circle 3 (NEW LOGIC)
if statusFlag == 1
    set(circles(3), 'FaceColor', [0 1 0])
else
    set(circles(3), 'FaceColor', [1 0 0])
end
%% ----- ARROW -----
latestThinfil = thinfil(:,end);
avgThinfil = mean(latestThinfil);
maxPressureExpected = 300;
scaledY = min(max(avgThinfil / maxPressureExpected, 0), 1);
new_y = 0.07 + (0.48 - 0.07) * scaledY;
set(arrow_ax, 'Position', [0.05 - arrow_w/2, new_y - arrow_h/2, arrow_w,
arrow_h]);
drawnow limitrate
end
end

```